



PHV018 V. 010E RELEASE NOTE

☒ **Adopt new version for all new tape out and all designs immediately**
☐ **Adopt new version for all designs, both on going and new start**
☐ **Adopt new version for new design start**
☐ **No action required**

1. LIBRARY INFORMATION

Library Name: *phv018*

Library Version: *010*

Release Note Version: *E*

Library Type: *Flash high voltage pad*

Library Status: *Silicon proven*

Technology: *0.18um, Embeded Flash Process*

Design Rule: *T-018-FH-DR-003 Rev 0.1(High Voltage and Flash Cell Portion)*

T-018-LO-DR-001 Rev 2.3

Spice Model: *T-018-FH-SP-001 Rev 0.2*

DRC Command File: *T-018-FH-DR-003-C1 V0.1b*

LVS Command File: *T-018-FH-SP-001-C1 V0.2a*

LVS_Calibre_0.18um_embflash_2p6m_1.8v+3.3v+hv.0.2a

2. UPDATE HISTORY

Date	Version	Changes	Customer action recommended
2003/4/17	010a	The first release version	
2003/10/17	010b	Update pin layer and prBoundary to match actual metal width of pad connection.	
2003/12/9	010c	Modify poly resistor width to match MM design rule	
2004/11/18	010d	Modified ESD protect circuit (cascade) to enhance the HV stress immunity.	
2004/12/16	010e	Remove redundant CAD layer 235 (FW)	

3. DESIGN KIT VERSION DEPENDENCY TABLE

The following table shows the valid combinations of design-kit versions

<i>rln</i>	<i>gds</i>	<i>gdf</i>	<i>Spi</i>	<i>apf</i>	<i>apt</i>	<i>sef</i>	<i>set</i>	<i>syn</i>	<i>vlg</i>	<i>vit</i>	<i>sdc</i>	<i>mdt</i>	<i>doc</i>	<i>Lpe</i>	<i>ibs</i>
010a	010a	010a	010a	010a	010a	010a		010a	010a				010a		
010b	010b	010b	010b	010b	010b	010b		010a	010a				010a		
010c	010c	010c	010c	010c	010c	010c		010a	010a				010a		
010d	010d	010d	010d	010d	010d	010d		010a	010a				010a		
010e	010e	010e	010d	010e	010e	010e		010a	010a				010a		

TSMC Flash LIBRARY

rln = release note	syn = Synopsys
gds = GDSII layout view	vlg = Verilog
gdf = GDSII phantom view	vit = Vital (VHDL)
spi = Spice/lvs netlist	sdc = Star-DC
apf = Apollo FRAM (phantom) view	mdt = Mentor DFT (Fastscan, DFT advisor)
apt = Apollo FRAM and CEL (layout) view	doc = Documents (Datasheets)
sef = Silicon Ensemble LEF (phantom) view	lpe = Layout Parasitic Extracted Spice Netlist
set = Silicon Ensemble LEF and Virtuso (layout) view	ibs = ibis model file

4. KNOWN PROBLEMS AND LIMITATIONS

- a. For the accurate function, the constant time is used in Verilog model and it will induce SFD backed-annotation errors. So, if the SFD is used, you can update the timing specs manually in the Verilog model and bypass the annotation errors.
- b. To prevent potential DRC errors, customers should place a placement blockage layer which is larger than flash memory and the layer should be large enough to prevent any layer of customer's database overlapping with the prBoundary (or FLASH) of flash memory.

5. SPECIAL NOTE

- a. DRC had errors for the min. poly area coverage. This check is for the whole chip poly density check. This violation can be waived in this cell DRC check.

---RULE CHECK RESULTS STATISTICS

RULECHECK PO.R.3 TOTAL Result Count = 1 (1)